

Problem Set 02 Discussion

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General Points

```
{r, error=TRUE}  
# Code that generates an error
```

- Great job on formatting and submitting your files!
 - Really nice writeups of thought processes
 - Good citations/mentions of Google/ChatGPT use, prompt inclusion
- Quarto won't render files with errors
 - Add `error=TRUE` to your code block header if it has an error but you don't want that error to interrupt execution of the rest of your notebook.
- Bonus points for everyone who figured out `group_by` and `summarize()` early!
- If you are unsure about a specific question or concept even if you got the code to work, flag it in the submission and/or talk to me later!

Vectors, Logicals, and Coercion

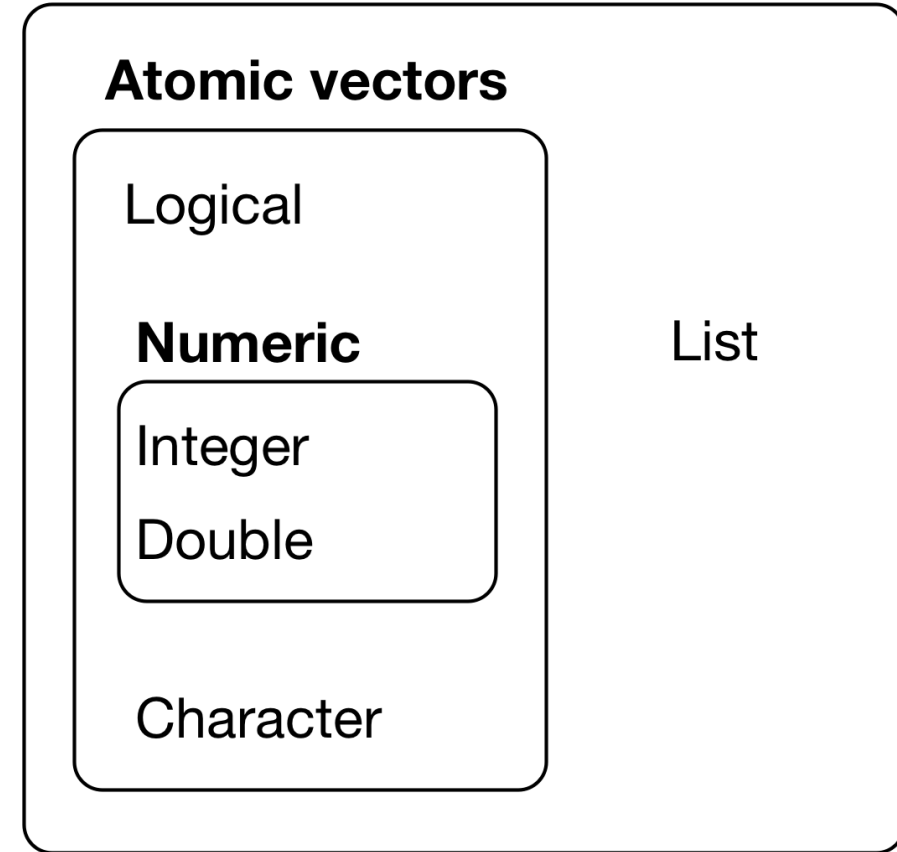
- General notes:
 - Logical = Boolean (0 or 1)
 - Logical values = `TRUE`, `FALSE`, `NA`

The `as.logical()` function attempts to convert other data types into logical values:

- **Numeric:** 0 becomes `FALSE`, any non-zero number becomes `TRUE`
- **Character:** Only `"TRUE"` and `"FALSE"` (case-sensitive) are converted; others become `NA`
- **Factors:** Converted to their underlying integer values, then to logical
- **NA:** Remains `NA`

logical -> integer -> double -> character

Vectors



NULL

Data Manipulation

- **study_sites** – named sublists

elevation habitat



```
study_sites <- list("windy_ridge" = list(2235, "grassland"),  
  "pine_valley" = list(1450, "forest"),  
  "stoney_creek" = list(900, "stream"),  
  "grassy_meadow" = list(1670, "grassland"),  
  "oak_glen" = list(1302, "forest"))
```

```
study_sites <- list(  
  "windy_ridge" = list(elevation = 2235, habitat = "grassland"),  
  "pine_valley" = list(elevation = 1450, habitat = "forest"),  
  "stoney_creek" = list(elevation = 900, habitat = "stream"),  
  "grassy_meadow" = list(elevation = 1670, habitat = "grassland"),  
  "oak_glen" = list(elevation = 1302, habitat = "forest")  
)
```

Can use multiple levels of \$ indexing!

```
study_sites$windy_ridge$elevation
```

Data Manipulation

- **study_sites** – named sublists

```
              elevation  habitat
              ↓         ↓
study_sites <- list("windy_ridge" = list(2235, "grassland"),
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```

Can use multiple levels of \$ indexing!

```
study_sites$windy_ridge$elevation
```

- **NC_births** – tricky unit conversion

- Imperial (pounds) to metric (kg)
- Could either divide by 2.205 or multiply by 0.4545
- father's age (fAge),
- mother's age (mAge),
- weeks of gestation (weeks)
- whether the birth was premature or full term (premature)
- number of OB/GYN visits (visits)
- mother's weight gained in pounds (gained)
- babies birth weight in pounds (weight)
- sex of the baby (sexBaby)
- whether the mother was a smoker (smoke).